

COMPLEMENTIZER - TRACE PHENOMENA AND THE NOMINATIVE ISLAND CONDITION*

DAVID PESETSKY

0.0 INTRODUCTION

My main goal in this paper will be to suggest a new explanation for what I shall call the *Complementizer-Trace Phenomenon* (CTP). This is the well-known constraint which, in many languages, prohibits WH-movement and other extractions from subject position in embedded clauses headed by a complementizer.¹ Thus, (1a) violates the constraint, while (1b), where the offending complementizer has been deleted, is acceptable:

- (1) a. *This is the guy who I said that ____ loved Mary
b. This is the guy who I said ____ loved Mary

The CTP has been much discussed in the recent literature. Chomsky & Lasnik (1977) have proposed a universal surface filter, which bars the sequence *complementizer-trace*, thus ruling out sentences like (1a). Bresnan (1977) has invoked a constraint on variables in transformations to account for the phenomenon, essentially ruling out extraction and deletion

* An earlier and shorter version of this paper appeared as Pesetsky (1979). A longer circulated draft contains much of the material found here. In revising the paper for somewhat belated publication, I have reworked a few sections under the influence of more recent work, but for a more substantial revision of some of these ideas see, *inter alia*, Chomsky (1981) and Bennis (1980). Important in this discussion was Freidin & Lasnik (1981), whose results, conflicting in some areas with those presented here, helped motivate the paradigm change suggested in Chomsky (1981). This paper itself owes a debt to an earlier version of Taraldsen (1980). For general comments and useful discussion I wish to thank particularly Joan Bresnan, Noam Chomsky, Ken Hale, Norbert Hornstein, Richard Kayne, S. Jay Keyser, Mark Liberman, Elena Padučeva, M.G. Sinicyn, Donca Steriade, Tarald Taraldsen and audiences at MIT, NELS IX and Moscow State University. This research was made possible by an NSF Graduate Fellowship and by a grant for research in the USSR from the International Research and Exchanges Board. All references refer to date of publication, where possible.

of an element over a variable ending in a complementizer. In both cases, as well as in the analysis of Taraldsen (1980), it is assumed that a special mechanism must be invoked to account for the phenomenon, which addresses itself specifically to the situation found in (1a). A recent analysis by Kayne (1980) is somewhat different in this regard, and I will discuss his proposal in Appendix I below.

In this paper I will show that the CTP can, in fact, be easily explained, without special mechanisms, within the general framework of the "Extended Standard Theory" including the *Nominative Island Condition* (NIC) of Chomsky (1980) and a minor reformulation of the filter on doubly-filled COMPs (cf. Chomsky & Lasnik, 1977). The only deviation required from the model of Chomsky & Lasnik and of Chomsky (1980) will be the ordering of the rule of Free Deletion in COMP before input into the interpretive components. I will suggest that such an ordering does not have any obvious ill consequences for the theory as a whole, and may, in fact, simplify it somewhat.

A complication will be necessary to account for certain problems with *that*-relatives in English and other languages, which have also given trouble to earlier analyses of the CTP. The solution I will suggest is interesting in that it parallels a reasonable solution for the French alternation between *que* and *qui*, and for complementizer alternations in other languages, such as Norwegian.

Once I have spelled out the mechanisms of the proposed explanation for the CTP, I hope to justify it by showing how it can deal with the problem of the CTP's universality. Even an examination of the surface distribution of the CTP among languages suggests that it has some universal basis. Chomsky & Lasnik have argued that it is not a learnable constraint, and therefore must arise from an innate mechanism. (See Phinney (1980) for some acquisition data.) This seems reasonable. In Russian, for example, extraction from embedded tensed clauses, contrary to some recent claims (Perlmutter, 1971; Chomsky, 1980), is possible, but has a rather marked status, and is best from subjunctive clauses. Such extraction is generally forbidden by the prescriptive grammarians (e.g., Rozental', 1970, cited by Zaliznjak and Padučeva, 1979), and occurs quite rarely in colloquial speech. Russian speakers surely have little or no exposure to these constructions in the crucial years of language acquisition. Nonetheless, informants have clear intuitions that the CTP holds:²

- (2) a. U menja est' kniga, kotoruju ja xoču, čtoby vy pročli
by me is book, which+ACC I wish, that+SUBJ you read
'I have a book which I wish you would read'
- b. *U menja est' kniga, kotoraja ja xoču, ctoby byla vo vsech bibliotekax
by me is book, which+NOM I wish' that+SUBJ was in all libraries

- 'I have a book which I wish would be in all libraries '
- c. Paren', kotorogo ja xotel, $\check{X}$ ioby Masa $\check{X}$ ubila ...
 guy, who+ACC I wished that+SUBJ Masha+NOM kill ...
 'The guy, who I wanted Masha to kill ...'
- d. *Paren', kotoryj ja xotel, $\check{X}$ toby ubil Ma $\check{X}$ u ...
 guy, who+NOM I wanted that+SUBJ killed Masha+ACC ...
 'The guy, who I wanted to kill Masha ...'

All the evidence suggests that Russians do not learn this restriction, but that it results from their innate linguistic capabilities.

Nonetheless, as Perlmutter points out, the CTP does not hold in all languages. In particular, he suggests that languages which show *null subjects* lack the CTP. For example, the constraint does not hold in Italian, as the following contrast between English and Italian shows:

- (3) a. *Who do you think that will come to visit?
 b. Chi credi che verra a visitarci?

This generalization appears to work across a wide variety of languages. Thus, any theory which seeks to explain the CTP and to account for its "universality" must also include an explanation for Perlmutter's generalization.

However, work by Maling & Zaenen (1978), and the Russian examples cited above somewhat cloud the clear picture suggested by Perlmutter's generalization. Maling & Zaenen show that, in some dialects of Dutch and in Icelandic, the CTP does not hold, even though these languages do not seem to show obvious cases of a null subject position. Russian, on the other hand, appears to lack subjects in many constructions, and yet shows the CTP. Clearly, if one is not to jettison Perlmutter's generalization, a theory of the CTP must not only incorporate the generalization, but also be able to draw on independently motivated facts about languages like Dutch, Icelandic and Russian to explain why they are exceptional with regard to it.

It should be obvious that the surface facts about the distribution of the CTP are rather complicated. To a certain extent this paper will necessarily suffer from being a Cook's tour through a number of languages and dialects. I will not be able to give an entirely adequate account of all the puzzles posed by the CTP in these languages; however, I hope to show that, although our theory will be motivated for English, it can come surprisingly close to explaining the range of phenomena we will find. In this connection, I will be comparing the present theory mainly with the universal surface filter proposed by Chomsky & Lasnik (1977), although I will also refer to other recent analyses.

Finally, I will suggest that the theory I present offers other dividends

besides an explanation of the CTP. In particular, it will naturally derive the major parts of Chomsky & Lasnik's **NP Tns VP* filter, which blocks deletion of a WH extracted from an adjacent subject position. The theory then makes a prediction: that languages which lack the CTP will also, if it is possible to tell, lack the **NP Tns VP* filter. This prediction, which is not made by other accounts of the CTP, appears to be correlated by data from Old English and Black English dialects. If the correlation is true, it provides strong support for the present analysis. It also seems to cast doubt on the entire system of universal and semi-universal surface syntactic filters developed by Chomsky & Lasnik, particularly in the light of Chomsky (1980)'s reanalysis of the phenomena previously accounted for by the **NP to VP* filter.

1.0 COMPLEMENTIZER-TRACE PHENOMENA

1.1. *Case Marking and the NIC*

For most of this paper, I will be working within the "Extended Standard Theory" (EST), as developed in recent work, assuming in particular WH-movement to a position in COMP, governed by the subjacency condition. Other possibilities are discussed in Appendix I below. Let us assume, with Chomsky (1980), that a general case-marking convention assigns the feature [+nom] to an NP governed by Tense. Other case-marking rules may assign other features to NPs, as in Chomsky (1980), but these are irrelevant for our purposes. It is stipulated that if this nominative NP is subsequently moved, as by WH-movement, the case marking remains on the moved NP and on the trace left at the original extraction site. Case is *not* left, however, on other traces created by the moved NP in the course of a derivation, as by successive-cyclic WH-movement through COMP.

Chomsky has proposed that a nominative "anaphor" is subject to a universal condition on binding, the *Nominative Island Condition* (NIC), which states:

A nominative anaphor cannot be free in $\bar{S}$

This condition replaces the Tensed S Condition of Chomsky (1973), and can explain the impossibility of reflexives, reciprocals and PRO as subjects of tensed sentences, since they are "anaphors" in the specialized sense of the NIC:

- (4) a. *John says that himself likes Mary
 b. *They all know that each other is a crook
 c. *I want PRO will come

As Chomsky notes, there is reason to suppose that the trace left by WH-movement is also an anaphor under the NIC. For example, French allows violations of the WH-island constraint, at least in some dialects (Sportiche, 1981). In these cases of multiple extraction of WH from S, a nominative WH cannot move past the nearest COMP:

- (5) a. Qu'est-ce que tu te demandes qui a donné à Marie?
 What you wonder who gave to Marie
 b. *Qui tu te demandes qu'est-ce qu'a donné à Marie?³
 Who do you wonder what gave to Marie

This contrast can be explained by assuming that WH-trace is subject to the NIC, given the following derived structures for (5a–b):

- (6) a. [$\bar{S}$ [COMP^{WH_j}] [S tu te demandes [$\bar{S}$ [COMP [^{WH_i}_{+nom}]]]]]
 [S [_{+nom}^{t_i}] a donné t_j à M.]]]]
 b. [$\bar{S}$ [COMP [_{+nom}^{WH_i}]] [S tu te demandes [$\bar{S}$ [COMP^{WH_j}]]]]
 [S [_{+nom}^{t_i}] a donné t_j à M.]]]]

In (6a), the nominative trace in the lower clause is bound by WH in its $\bar{S}$, and thus does not violate the NIC. In (5b), however, the nominative trace is not bound in its $\bar{S}$, and does violate the NIC. I follow Sportiche's analysis, under which a choice of S as bounding for Subjacency permits WH-movement to skip the COMP nearest its extraction site in S, deriving structures like (5a–b). (See also Rizzi, 1978, for the original statement of this analysis, for Italian.) Note that WH is not an "anaphor" under the NIC; thus the nominative WH's in COMP in (6a–b) do not violate the condition.

Thus, data like (6a–b) suggest that trace, at least in French, is an anaphor under the NIC. Note that a contrast parallel to (5a–b) is found in English as well, despite the oddity of the WH-island violation:

- (7) a. ?This is one crime that I know what criminal committed
 b. *This is one criminal that I know what crimes committed

Thus, it looks as if trace is an anaphor under the NIC in English, too.

1.2. Proposal

We are now ready to consider how the NIC can explain the CTP, the phenomenon exemplified by (1a–b) above. We want to account for the fact

that extraction from subject position in an embedded clause is ungrammatical if that clause begins with a complementizer, while deletion of the complementizer renders the extraction grammatical. As Bresnan (1977) has noted, the CTP is not limited to the environment of the complementizer *that*, but generalizes to all complementizers, or at least to those which begin tensed sentences. (She argues for the inclusion of *for* in the generalization; I discuss this in section 5.2 below.) Thus:

- (8) a. *This is one guy who I don't know if ___ likes Felicia
 b. *The man I asked whether — had seen the film was Joe

Since *whether* and *if* are undeletable, there are no grammatical sentences parallel to (6a–b).

Now let us consider the deep structure of the relevant NP in (1a–b) after case marking has applied:

- (9) the guy [$\bar{S}$ [COMP₁ that] [S I said [$\bar{S}$ [COMP₂ that]
 WH_i
 [S [+nom] loved Mary]]]]

WH-movement applies, and nominative case is left on the trace in the lowest S, as described above:

- (10) the guy [$\bar{S}$ [COMP₁ WH_i
 $[+nom]$ that] [S I said [$\bar{S}$ [COMP₂ t_i that]
 t_i
 [S [+nom] loved Mary]]]]

The nominative trace in S is bound by the trace left in COMP₂, so, as it stands, (10) is not in violation of the NIC. It is clear that if this trace is deleted before application of the NIC the resulting sentence *will* violate the NIC. We want a condition which will force us to delete this trace only when it shares a COMP node with an undeleted complementizer.

Fortunately, there is an already motivated condition which will do just this. We are assuming here a model in which WH moves into a slot in COMP. If COMP can also contain a base-generated complementizer, we need a condition to prevent bad sentences like (11):

- (11) *The man who that likes Mary is Bill

Chomsky & Lasnik, following a suggestion by Keyser (1975), suggest a surface filter to cover this. They state it thus:

- (12) *[COMP *wh*-phrase φ], $\varphi \neq e$ (=C&L (53))

Filter (12) is bled (preceded) by a rule of *Free Deletion in COMP*, which can eliminate what would be violations of the filter.

Suppose we extend this filter to include traces of WH-phrases as well as WH itself:

$$(13) \quad *[\text{COMP} \left\{ \begin{matrix} \text{WH} \\ t \end{matrix} \right\} \varphi] \dots$$

We can state this as a much more general condition, as in (14):

$$(14) \quad *[\text{COMP} \alpha \beta] \quad (\alpha \text{ and } \beta \text{ immediately dominated by COMP})$$

If (14) is correct, then, in order to yield a well-formed surface string, the rule of *Free Deletion in COMP* will have to apply in both COMP_1 and COMP_2 . For present purposes, what happens in COMP_1 is irrelevant. Let us say that the complementizer *that* is deleted, as in (1a–b). What is interesting is COMP_2 . Here either the complementizer or the trace must delete, in order not to violate (14). The two possible outputs are (1a–b), whose derived structures are given in (15):

$$(15) \quad \begin{aligned} \text{a. the guy } & [\bar{S} [\text{COMP}_1 \overset{\text{WH}_i}{[+_{\text{nom}}]}]] [\text{S I said } [\bar{S} [\text{COMP}_2 \text{ that }] \\ & [[\text{S } [{}^t_{+_{\text{nom}}}] \text{ loved Mary }]]]] \\ \text{b. the guy } & [\bar{S} [\text{COMP}_1 \overset{\text{WH}_i}{[+_{\text{nom}}]}]] [\text{S I said } [\bar{S} [\text{COMP}_2 t_i] \\ & [[\text{S } [{}^t_{+_{\text{nom}}}] \text{ loved Mary }]]]] \end{aligned}$$

In (15a), where trace was deleted in COMP_2 , the nominative trace in the lower S is now unbound in $\bar{S}$ and is in violation of the NIC, correctly ruling the sentence out. In (15b), the trace has not been deleted in COMP_2 , and the sentence, as predicted, is not ruled out by the NIC, since this trace binds the nominative trace in $\bar{S}$.

The filter prohibiting doubly-filled COMPs will, of course, always motivate the deletion of trace in COMP when it is the sister of an undeletable complementizer like *if* or *whether*. Our analysis thus gives the full range of effects of the CTP, relying almost entirely on independently motivated theory.

1.3. *Non-nominatives and the Opacity Condition*

Our analysis will not falsely exclude grammatical sentences like (16), where a non-nominative WH has been extracted from a clause headed by a complementizer:

- (16) This is the guy who I said that Mary loved

This has the derived structure (after deletion from COMP):

- (17) ...the guy [$\bar{S}$ [COMP [$\overset{WH_i}{-nom}$]] [S I said [$\bar{S}$ [COMP that]
[S Mary loved [$\overset{t_i}{-nom}$]]]]]

The trace in the lower S is not bound in its $\bar{S}$, but the NIC is silent about non-nominative anaphors.

Note, however, that if WH-trace is an anaphor under the NIC, it cannot be subject to the Opacity Condition, Chomsky's (1980) revision of the Specified Subject Condition. In (17), after deletion in COMP, the lower S contains a trace which is free in the domain of a subject. This should violate Opacity if trace is an "anaphor" under this condition. Nonetheless, the sentence is acceptable. I have nothing to say on this subject here.⁴ Note that if Rizzi's (1978) analysis of WH-island violations is correct (and cf. (6a–b) for French), he too must assume that Opacity does not apply to WH trace.

1.4. *Ordering of Components*

Contrary to the assumptions of Chomsky & Lasnik (1977) and Chomsky (1980), the present proposal requires that the deletion rule in COMP precede semantic interpretation, where the NIC applies. Chomsky & Lasnik suggest that deletion rules in general must follow semantic interpretation, "since deleted elements must undergo semantic interpretation". For lack of evidence to the contrary, I will suggest that the opposite ordering is not impossible. It is, in fact, somewhat suggested by the need, in Chomsky & Lasnik's theory, for a "Recoverability Condition" on deletion. This condition, which has never been precisely formulated, to my knowledge, is invoked to account, among things, for the non-deletability of a pied-piped PP in COMP. The claim is often made that the preposition of such a PP has some sort of "lexical status" absent from an element like WH. If this is a true explanation, a serious "autonomous systems" approach to linguistic theory would have some difficulty with the conditioning of a syntactic deletion rule by a seemingly semantic factor like "lexical status". (Alternatives are, of course, available.) In the case of deletion, a satisfactory account of the restriction on PP deletion in COMP could presumably be obtained by allowing deletion to operate freely, before semantic interpretation. The deletion would then be required to be recoverable, literally, by semantic recovery rules deriving Logical Form, or the sentence would be ruled out by a general restriction on uninterpreted nodes or on unfilled argument positions. (See also Woolford, 1978,

for arguments in favor of ordering deletions before interpretation and for doing without any Recoverability Condition, at least for PP deletion in COMP.

The foregoing remarks are obviously speculative. I merely wish to show that there are no obvious and strong arguments against ordering deletions before semantic interpretation. If the present analysis is correct, it will, of course, provide an argument in favor of this ordering.

2.0 PROBLEMS WITH RELATIVE CLAUSES IN ENGLISH, NORWEGIAN AND FRENCH

2.1. *That-relatives in English*

The analysis so far has been shown to account adequately for the CTP in English, and we shall assume that it represents the right explanation for the CTP in other languages in the sections to come. It runs into problems, however, with extractions which stop in the nearest COMP. These problems parallel those which other treatments of the CTP have come up against.

Consider an underlying structure for an NP as in (18), after case marking:

$$(18) \quad [\text{NP the man } [\bar{S} [\text{COMP that}] [S [\text{WH}_i \text{ } +\text{nom}] \text{ liked spinach }]]]]$$

WH-movement will yield (19):

$$(19) \quad [\text{NP the man } [\bar{S} [\text{COMP } [\text{WH}_i \text{ } +\text{nom}] \text{ that}] [S [\text{t}_i \text{ } +\text{nom}] \text{ -liked spinach }]]]]$$

At this point, our filter against doubly-filled COMPs will force us to use the rule of Free Deletion in COMP to delete either *that*, WH or (optionally) both from COMP. If we delete just the complementizer *that*, we have the good output:

$$(20) \quad [\text{NP the man } [\bar{S} [\text{COMP } [\text{WH}_i \text{ } +\text{nom}]] [S [\text{t}_i \text{ } +\text{nom}] \text{ liked spinach }]]]]$$

If we delete both WH and *that* in COMP, the result, with the structure given, will be ruled out by the NIC, since the trace remaining in subject position will be nominative and free in $\bar{S}$:

$$(21) \quad *[\text{NP the man } [\bar{S} [\text{COMP } e] [S [\text{t}_i \text{ } +\text{nom}] \text{ liked spinach }]]]]$$

This is an important result of this analysis, since this phenomenon must otherwise be handled by an ad hoc mechanism, such as Chomsky &

Lasnik's *NP *Tns VP* filter. We will return to this point in section 4 below.

If we delete only the WH, however, we have trouble. Under the analysis presented so far, the WH is the only binder of its nominative trace in S. Logically, its deletion should produce a violation of the NIC. The result, however, is grammatical:

- (22) [NP the man [$\bar{S}$ [COMP that] [S [t_i _{+nom}] liked spinach]]]

This construction has been a problem for earlier theories of the CTP, and has required special stipulation. For example, Chomsky & Lasnik account for the CTP by a surface filter, whose core is:

- (23) * [$\bar{S}$ $\pm$ WH [NP^e] ...]

(Where $\pm$ WH is any complementizer except *for*, in English.) It is clear that this filter will also incorrectly rule out (20), so Chomsky & Lasnik add a proviso:

- (24) ...unless $\bar{S}$ or its trace is in the context [NP NP — ...] (= C&L (85))

In our theory, the CTP is accounted for by a condition on binding. Thus our “out” to allow (20) must itself involve binding. I will claim that the “out” of choice, while not independently motivated in English, is so motivated in other languages where the same problem presents itself.

2.2. COMP Contraction

The solution I will adopt, although other formulations are possible, involves a “contraction” rule applying in COMP before semantic interpretation. Clearly, if an appropriate mechanism can transfer the referential index from a WH in COMP to a sister complementizer, the correct pattern of data will be generated. (22) would not be ruled out by the NIC, since the nominative trace in S will be bound in $\bar{S}$ by the complementizer. (21), however, will still be correctly ruled out, since deletion of both complementizer and WH will leave nothing at all binding the trace in $\bar{S}$.

Let us state this rule as a special case of deletion in COMP, thus:

- (25) [COMP WH_i complementizer] → [COMP complementizer_i]⁵

Applying this rule to (19), we get a string homophonous with (20), but not ruled out by the NIC;

- (26) [NP the man [$\bar{S}$ [COMP that_i] [S [t_i _{+nom}] liked spinach]]]

Note that (25) crucially does *not* apply to WH-*trace*, or it would permit violations of the CTP in English.

As it stands, the rule proposed is an ad hoc solution to the problem of (22). It is important, therefore, that mechanisms like (25) can be independently motivated in other languages, where they are correlated with a phonological effect. What (23) claims, in essence, is that there is something special about the complementizer which introduces relative clauses, whereby it is distinct from a neutral, non-relative complementizer.⁶ This is obviously true in some languages.

2.2.1 Norwegian

In Norwegian, the neutral complementizers *at* and *som* are in complementary distribution. *At* introduces non-relative subordinate clauses as an equivalent of English *that*, but never introduces relative clauses. *Som*, on the other hand, only introduces relative clauses (with all the familiar preposition-stranding properties of English *that*). Following Taraldsen (1980), we note that Norwegian acts like English with respect to the CTP. Assuming the analysis presented here, Norwegian will need a mechanism to account for the possibility of *som*-relatives with extraction from a nominative position, as in (27a–d), parallel to (19, 21–22) above:

- (27) a. [NP en Mann [$\bar{S}$ [COMP WH_i at] [S [NP ^{t_i}_{+nom}] var fra India]]]
 b. *[NP en Mann [$\bar{S}$ [COMP ^e] [S [NP ^{t_i}_{+nom}] var fra India]]]
 c. *[NP en Mann [$\bar{S}$ [COMP at] [S [NP ^{t_i}_{+nom}] var fra India]]]
 d. en Mann som var fra India ...
 ‘a man that was from India ...’

It turns out that a rule analogous to (25) will do the trick, and account for the *som/at* alternation at the same time.

- (28) [COMP WH_i at] → [COMP som_i]

What is reassuring about (28) is that a visible morphological change is brought about by the rule, and it appears somewhat less of a “magical” rule than (25).

2.2.2 French *que/qui*

A similar, but more complex rule seems to account nicely for the alternation of *que* with *qui* in French, and for certain puzzling facts about its interaction with the CTP.⁷

With the neutral complementizer *que*, French shows the CTP. Extrac-

tion of a subject from a clause headed by *que* is ungrammatical. For many speakers, however, the extraction is grammatical if *que* is changed to *qui*. (This observation is originally due to Perlmutter, 1971.) Thus:

- (29) a. *L'homme *que* tu crois *que* ____ viendra nous rendre visite...
 The man that you believe that will come pay us a visit...
 b. L'homme *que* tu crois *qui* ____ viendra nous rendre visite...

Let us explain (29a) as we have explained the CTP in English. To explain (29b), we need to say that *qui*, introducing a non-relative clause, has a referential index which binds the nominative trace in $\bar{S}$ and prevents an NIC violation. *Qui* does not occur freely as a complementizer, but only "when needed" to avoid an NIC violation. Whatever rule governs its occurrence must mention binding of a nominative position. Thus:

- (30) a. L'homme *que* tu crois *que* j'aime ____ ...
 The man that you believe that I like ...
 b. *L'homme *que* tu crois *qui* j'aime ____ ...

Now note that *qui* will also occur in our troublesome *that*-relative case, in a simple relative clause with extraction from a nominative subject position:

- (31) a. *L'homme *que* ____ viendra nous rendre visite ...
 the man that will come pay us a visit ...
 b. L'homme *qui* ____ viendra nous rendre visite ...

In this context, too, *qui* will only occur "when needed". The generalization, given the present analysis, is clear. *Qui* occurs instead of *que* when, before deletion in COMP, a WH or trace adjacent to *que* in COMP binds a nominative trace in the sister S. In other words, *qui* is a form of *que* which provides an "escape hatch" from the effects of the NIC. A rule governing this will resemble (25) and (28) except for the inclusion of trace and the specification of a binding environment. It may be possible to restate the binding environment as a condition on *qui* in Logical Form. For our immediate purposes, however, we may state the rule as (32):

$$(32) \quad \text{WH}_i \text{ que } \rightarrow \text{[COMP } t_i \text{ qui}_i \text{]} / \text{ ____ [S x [} t_i \text{ }_{+nom} \text{] y]}^8$$

For some French speakers, the change of *que* to *qui* does not provide an escape from the CTP, while it does act in simple relative clauses as in (31). In these dialects, rule (32) simply does not mention trace on the left side of the arrow, bringing it closer to the rules for English and Norwegian. Notice

that the English rule could be written (without the inclusion of trace to the left of the arrow) like the French rule, since English only needs the rule for subject relatives. If we suppose that these rules are language-specific, in content if not in form, however, then one must suppose that a learner of English will develop the maximally simple rule to cover *that*-relatives, which will resemble the Norwegian rule more than the French rule.

Of course, we cannot claim that the "contraction rule" approach will ultimately prove to be the best solution to these problems from a theoretical point of view. Other approaches, such as base generation of the relevant complementizers, subject to interpretive conditions, are imaginable. My main wish has been to show, with the French and Norwegian examples, that the problem with sentences like (22) is not an artifact of the present analysis, but reflects a real feature of UG whose solution will have ramifications for what are obvious problems in other languages.

3.0 NULL SUBJECTS AND THE NIC

3.1. *Perlmutter's Generalization*

Questions of learnability and generality, discussed in the Introduction, have suggested that the CTP has some sort of universal basis. As I have said, Perlmutter noticed that languages which independently allow null subjects, like Italian and Spanish, do not in general show the CTP. Thus (33a–b) in Italian contrast with (34a–b) in English:

- (33) a. Parlo italiano
 b. Chi credi che ____ verria a visitarci?
- (34) a. *Speak Italian (= I speak Italian)
 b. *Who do you think that ____ will come to visit?

Clearly some parameter in the present theory must account for this difference. In our theory, the universality, such as it is, of the CTP is a result of the universality of the NIC. One must therefore look for appropriate language-particular parameters which will allow the CTP to be violated in a given language without running afoul of the NIC.

One possible parameter would be a language's lack of the filter against doubly-filled COMPs, (14). I have found no clear cases in which CTP violations can be blamed on the lack of this filter (see section 6.0 below), although I have found no counterexamples to this prediction either.⁹ In any case, this parameter would not help us to explain Perlmutter's generalization.

We must therefore look for some parameter along which languages can

vary with respect to the NIC itself. Let us assume that this parameter is an allowable variation among languages in the items that are “anaphors” with respect to the NIC.¹⁰ We will suppose that what we have been calling “trace” is simply a co-indexed null NP, [_{NP}e], interpreted as a variable by rules of Logical Form. Let us suppose that languages can differ in whether null NP is an “anaphor” for the NIC. If the NIC applies to trace it will also apply to other null NPs; if it does not apply to trace it will also not apply to other null NPs. A “null subject” is simply a nominative instance of null NP (in the languages we have discussed so far; see below on Russian), and is thus ruled out in English and in other languages where the NIC applies to null NPs, like French and Norwegian. In languages like Spanish and Italian, null subjects are possible, because null NP is not an anaphor for the NIC. It is clear that if null NP is not an anaphor for the NIC, the language will violate the CTP, since nominative traces will not have to be bound in $\bar{S}$.

This makes the further prediction that the restrictions on WH-island violations found in French (section 1.1 above), which we attributed to the application of the NIC to trace, will not be found in Italian. This prediction is borne out, as Luigi Rizzi (1978) notes. Thus French (35a) contrasts with Italian (35b):

- (35) a. *L'homme que tu te demandes qu'est-ce qu'a donné à Marie ...
 b. La persona che ti domandi che cosa ha dato a Maria ...
 The person that you wonder what gave to Mary

That Italian should differ from French and English in precisely these cases I take to be good evidence for the present approach. Kayne (1980) provides further possible examples of the effect of this parameter on inter-language variation.

Note that we cannot say that the NIC is simply not valid at all in null-subject languages, since most such languages with which I am familiar preserve the constraint against nominative reflexives, reciprocals and PRO. Why null NP should be a parameter for variability, while reflexives and reciprocals should not, is a question I cannot answer. (But see Chomsky, 1980, for some remarks on “reflexives” in Japanese and Korean that may not be subject to the NIC.)

Finally, note that the sort of parameter I am suggesting here does not raise any learnability problems. In general, we can suppose that if a child encounters CTP violations or null subjects, he or she will conclude that null NP is not an anaphor for the NIC in that language. (We will need to be somewhat more precise about the meaning of the term “null subject” when we consider Russian below.)¹¹

3.2. *Dutch and Icelandic*¹²

Chomsky & Lasnik, assuming the filter of (23—4) above, have accounted for Perlmutter's generalization by claiming that null-subject languages have a rule deleting pronominal subjects, including trace. Since this rule precedes their filter, potentially offending occurrences of [_{NP ϵ}] are deleted before the filter applies, allowing them to escape the CTP. In an interesting paper, Maling & Zaenen (1978) have presented evidence from Icelandic and Dutch which causes some trouble for this solution. According to their data, Icelandic and at least one dialect of Dutch do not generally allow null subjects, but show violations of the CTP. They take this as evidence against the universality of the filter, since no plausibly general rule of Subject Pro Drop seems to obtain in these languages. Note that their argument does not show that Chomsky & Lasnik's framework is *incapable* of handling the Dutch and Icelandic data. It merely shows that one would need an ad hoc deletion rule bleeding the filter — one which specifically mentioned [_{NP ϵ}]. I shall claim, however, that a more interesting solution is available in the framework of this paper.

I should note at the outset a plausible non-solution for the Dutch data we will examine. Annie Zaenen has pointed out to me that some Dutch dialects show constructions with what appear to be doubly-filled COMPs. In our framework, this could explain the presence of CTP violations. Since this is apparently not true in as many dialects as allow violations of the CTP, I will assume that this is not a primary explanation of Dutch CTP violations.

Let us begin with the minimal hypothesis about the analysis of CTP data. I will assume that if a language lacks doubly-filled COMPs (*pace* the preceding remarks) but shows CTP violations, then null NP is not an anaphor for the NIC in that language. Thus, the problem in Dutch and Icelandic will *not* be to account for violations of the CTP, but rather to account for the absence, in most cases, of null subjects. Neither language appears to allow empty COMPs; they have a common filter against [_{COMP ϵ}]. Therefore we do not expect to find violations of the *NP *Tns* VP filter of Chomsky & Lasnik parallel to (21) above.

I will concentrate on Dutch, since, according to Maling & Zaenen, Icelandic acts in much the same way. Perlmutter first noted that CTP violations are found in Dutch, as shown in (36):

- (36) Wie vertelde je, dat — gekomen was?
 Who said you, that come had

It appears that the acceptability of (36) is limited to certain dialects. Those dialects which accept sentences like (36) we will call, following Perlmutter, Dutch A, while those which do not we will call Dutch B. Let us

assume that the essential difference between Dutch A and Dutch B is that, in the former dialect, null NP is not subject to the NIC, while in Dutch B it is.

Now if the NIC does apply to null NP in a given language, we predict that *no* cases of null nominative subjects will be found. This is precisely the case in Dutch B. If in a given language the NIC does *not* apply to null NP, we conversely expect that null subjects will be possible. In fact, they are possible in Dutch A, but with a very limited distribution. In particular, they are not possible, as they are in Italian, with interpretation as a definite personal pronoun. Since this sort of null subject is a salient phenomenon in the surface data of a language, there is a tendency to use it as a defining characteristic of a "null subject language". I will suggest that this is a false move.

Maling & Zaenen discuss the principal dummy subject in Dutch, *er*, whose distribution is similar to, but wider than that of English *there*. It is found in sentences analogous to English existentials and presentational *there* sentences, and also fills subject position (perhaps) in impersonal passives. What is crucial is that *er*-insertion is optional in Dutch A, wherever its non-application will not violate the verb-second (V/2) constraint – for example, when another fronting rule has applied or in an embedded clause. In Dutch B, however, *er*-insertion (whether transformational or lexical) is always obligatory. Thus, the impersonal passive shown in (37), with a rightwardly moved subject, is grammatical with *er* in subject position in both dialects. Its counterpart without *er*, (37b), is, however, grammatical only in Dutch A:

- (37) a. Gisteren werd er door het hele dorp gedanst en gezongen
 Yesterday was there by the whole village danced & sung
 (= M&Z (31))
 b. Gisteren werd door het hele dorp gedanst en gezongen
 (=M&Z (31))

Thus, with regard to these data, the present analysis makes the right predictions about the two dialects of Dutch. In Dutch A, null NP is not subject to the NIC, and we find NIC violations and null subjects. In Dutch B, null NP is subject to the NIC, and we find neither. We must still explain why Dutch A does not allow null *personal* subjects, that is, null subjects interpreted as personal pronouns. Thus, (38b) is not a grammatical paraphrase of (38a) in any dialect:

- (38) a. Hans vertelde, dat hij gekomen was
 Hans said that he come had
 b.*Hans vertelde, dat gekomen was

It has been pointed out, in other contexts, by Taraldsen (1980) and by Kayne (1980), that the possibility of sentences like (38b) does not rest on non-application of the NIC alone. It is not enough simply to say that a given language allows null subjects in tensed sentences. This will only suffice when the subject position does not require semantic interpretation, as in the *er*-insertion cases.¹³ If a null subject must be interpreted as a personal pronoun, however, a special rule of interpretation is necessary. This rule must "fill in" necessary information about the nature of the subject, such as person and number. Some of this is typically read off the verbal morphology. Clearly, the morphology of the Dutch verb is not rich enough to feed such an interpretive rule. (Of course, as we shall see, rich verbal morphology is perhaps a necessary, but clearly not a sufficient criterion for null subjects.) Note that a real interpretive rule is necessary, since some information, such as the definiteness of the subject (in Italian and similar languages) must be specified by it.

I will not try to further characterize the subject interpretation rule here. I only wish to suggest that all one must say about Dutch to explain the contrast between (37) and (38) is that it lacks this one rule. One might suppose that the presence of the rule is the marked case, although this has statistical problems, as discussed in footnote 11. A language-learner will not assume that such a rule exists until given evidence of its necessity. In Dutch, perhaps because of its impoverished verbal morphology, the rule does not exist. Thus, although Dutch A also allows null subjects in general, since they do not violate the NIC, it will not allow null personal subjects, since these are uninterpretable.

One problem with this analysis is the dummy *het* (cf. English *it*) used in "weather" constructions and in a variety of $\bar{S}$ extraposition contexts. Maling & Zaenen claim that *het* can never be deleted in any position. Their examples, however, all involve main clauses, where the V/2 restriction may render opaque the reasons for the obligatoriness of *het*. If we consider embedded clauses, however, a different picture emerges. "Weather" *het* is, as Maling & Zaenen say, undeletable:

- (39) Hij vertelde, dat *(het) regent
He said that it rains

However, as M.A.C. Huybregts (personal communication) has pointed out, *het* extraposition contexts *is* deletable with many predicates:

- (40) a. toen (het) bleek dat hij er was ...
when it appeared that he there was ...

- b. hij vertelde dat (het) duidelijk is, dat hij er was¹⁴
he said that it clear is that he there was

Het is not deletable with certain verbs, like *schijnen* 'seem' (41)a and with verbs that take auxiliary *hebben* 'have', for some speakers (like *verbazen*, (41)b):

- (41) a. toen *(het) scheen dat hij er was ...
 when it seemed that he there was ...
- b. hij vertelde dat %(het) hem zou verbazen als...
 he said that it him would surprise if...

While we have no explanation for the data in (41), (40) clearly suggests that, as predicted, empty subjects may replace the dummy *het* in Dutch. As for the "weather" constructions of (39), we must assume that the subjects of weather verbs are not complete dummies, but require some special interpretation (cf. the treatment in Chomsky (1981), where they are called "quasi-arguments").

Icelandic seems to admit a solution parallel to that proposed for Dutch. I will give only the main points.

Like Dutch A, Icelandic does not allow null personal subjects, but shows violations of the CTP. Here, as Maling & Zaenen point out, there is no dialect variation:

- (42) þetta er maðurinn, sem þeir halda að — sé of heimskur til
 his is man-the that they think that is too dumb in
- að vinna verkið.
 order to do job-the (= M&Z (17))

Icelandic strictly observes a V/2 constraint in both root and embedded contexts, unlike Dutch, A dummy *það* occurs, but we cannot look to see if it is optional or obligatory when moved to the right, since it undergoes no movement rules. Indeed, as Maling & Zaenen note, *það* seems to exist merely to satisfy the V/2 constraint. If any fronting rule applies, or if the subject of the sentence is not rightwardly moved, *það* is not inserted. If we can view such fronting rules as movement into subject position, we could say that the NIC, whether or not it applies to null NP, is not violated. The nominative NP slot will be filled, even if not by an NP. We will need to stipulate a similar analysis for some cases of PP-fronting in English (see 5.1 below). If, however, fronting does not fill subject position, we will be able to motivate a case for a null subject position in the impersonal passive (43):

- (43) Í gær var dansað (= M&Z (53a))
 Yesterday was danced

það does not occur, because the fronting of the adverb satisfies the V/2 constraint. If *I gær* is not in subject position, then clearly the subject position must be empty, and null NP will have to be exempt from the NIC, as we require for (42), to allow (43).

Thus, it appears that either Icelandic allows null, non-personal subjects, or it provides no relevant evidence, because of the V/2 constraint, which will motivate movement into subject position if it is left vacant.

As for null personal subjects, these do not appear to occur. (44b) is not possible as a parallel to (43). (We have postposed the subject so that it will not be crucially necessary to satisfy the V/2 constraint.)

- (44) a. *Í gær var mikill snjór á jörðinni* (= M&Z (67c))
 yesterday was much snow on ground-the
 b. **Í gær var á jörðinni*
 yesterday was (it, he, she...?) on ground-the

We will account for this as we did in Dutch A, by saying that Icelandic lacks the appropriate interpretive rule.

Since we have independently accounted for the non-existence or limited distribution of null nominative subjects in Icelandic, we can claim without any problems that CTP violations like (42) are due to the immunity of null NP to the NIC.

3.3. *Russian*

As noted in the introduction, Russian shows the CTP, even though the relevant extractions are often stylistically marked (see again Zaliznjak & Paduceva, 1979, and Chvany, 1975). We should emphasize, as discussed above, that there is a clear contrast for native speakers between extraction from an indicative clause and from a subjunctive clause. A nice minimal pair is:

- (45) a. **Paren', kotorogo ja skazal, čto Ivan ubil __*, sidit na divane
 Guy, who+ACC I said, that Ivan killed __, sits on couch
 b. *Paren', kotorogo ja skazal, čtoby Ivan ubil __*, sidit na divane
 Guy, who I said, that+SUBJ Ivan kill ('that I, should kill')...

We can see the CTP in Russian in examples like (2b, d) above, or (46), parallel to (45):

- (46) **Paren', kotoryj ja skazal, čtoby __ ubil Ivana, sidit na divane*
 Guy who+NOM I said, that+SUBJ kill Ivan+ACC, sits on couch
 'The guy who I said should kill Ivan is sitting on the couch'

Russian provides an impressive argument that the CTP cannot be properly explained by Chomsky & Lasnik's filter (23–4) or by any theory which, like Bresnan's Complementizer Constraint on Variables or the theory of Taraldsen (1980), takes adjacency of WH to COMP as a crucial cause of the phenomenon. Unlike English, Russian allows a good deal of scrambling, although SVO order is clearly unmarked. I leave open the question of whether scrambled word order is transformationally derived or base-generated. There are some interesting and important constraints on scrambling from within major constituents, but I will not discuss them here, since they seem irrelevant. One can show, however, that scrambled word order must be established before a Chomsky-Lasnik-style grammar forks – that is, before input into the semantic interpretive component. In other words, scrambling cannot be stylistic. Crucially, the order of constituents affects quantifier scope, much as in English. Thus, the following sentences differ in unmarked interpretation, as indicated by the English paraphrase and logical representation below. (Both Russian sentences are active.)

- (47) a. Mnogie profesora znajut vsech studentov
 [Many professors]+NOM know [all students]+ACC
 'Many professors know all the students'
 $Mx: x \text{ professor } (\forall y: y \text{ student } (Kxy))$
 b. Vsech studentov znajut mnogie profesora
 [All students]+ACC know [many professors]+NOM
 'All the students are known by many professors'
 $\forall y: y \text{ student } (Mx: x \text{ professor } (Kxy))$

Since scrambled order is syntactically determined, if rules interpreting quantifiers and the NIC are both in the "semantic" ('right') side of a Chomsky-Lasnik grammar -- the reasonable assumption -- scrambled order must be created before the NIC applies. In Chomsky & Lasnik's system, too, scrambled order must be created before semantic interpretation, and thus before filters.

Scrambled order can occur in embedded contexts; it is not a root phenomenon:

- (48) Ivan skazal, čto vsech studentov znajut mnogie profesora zdes'
 Ivan said that [all students]+ACC know [many professors]+NOM
 here

There is also no particular problem with extraction from embedded clauses with scrambled order, or with fronted constituents:

- (49) Paren', kotorogo ja xotel, čtoby v sentjabre ubil Ivan ...
 Guy, who+ACC I wished that in September kill+SUBJ Ivan+NOM
 'The guy who I wanted Ivan to kill in September ...'

We can now demonstrate that Chomsky & Lasnik's surface filter (and, incidentally, Bresnan's and Taraldsen's formulations) is too weak to account for the CTP in Russian.

The point is, there should always be a derivation for a CTP violation in Russian in which the extraction site is not adjacent to COMP, and thus there should always be an escape route from a filter like Chomsky & Lasnik's. Thus, in (50a) both subject and object can follow the verb, while in (50b) the object can precede the subject:

- (50) a. V sentjabre ubil Ivan Mašu
 In September killed Ivan+NOM Masha+ACC
 b. Mašu Ivan ljubil
 Masha+ACC Ivan+NOM loved

Extraction from the site of nominative *Ivan* in these sentences should in neither case violate Chomsky & Lasnik's filter. Under our theory, it will leave the nominative null NP unbound in $\bar{S}$, and should be ungrammatical. The facts support our theory:¹⁵

- (51) a. *Paren', kotoryj ja xotel, čtoby v sentjabre ubil t Masu ...
 Guy who+NOM I wanted that in September kill+SUBJ t Masha+ACC
 b. *Paren', kotoryj ja xotel, čtoby Mašu t ljubil ...
 Guy who+NOM I wanted that Masha+ACC love+SUBJ

We take this as strong evidence in favor of a case-related theory of the CTP over any positional theory, like Chomsky & Lasnik's. Additional proof that the CTP is to be accounted for in terms of case marking is provided by Russian existential sentences. We discuss below the problems raised by the lack of a pleonastic subject here. For now, note that the "focus" of these sentences, the quantified NP, is generally found to the right of the verb in both positive and negative existentials.¹⁶ A peculiarity of this construction is the fact that the focus of a positive existential is always in the nominative case, while the focus of a negative existential is in the genitive (like the object of a negative transitive verb. see Babby, 1978, 1980; Timberlake, 1975 for some discussion; also Pesetsky and Sinicyn (forthcoming) for an analysis within the EST framework). Thus:

- (52) a. Na stole byla kniga
 On table was book+NOM 'There was a book on the table'
 b. Na stole ne bylo knigi
 On table not was book+GEN 'There wasn't a book on the

There is surely no obvious semantic or relational difference between the focus of a positive and of a negative existential, nor is there any necessary positional difference. The only difference lies in the case marking. (Nominative case in (52a) triggers gender agreement in the verb; genitive case in (52b) renders the verb neuter.) Theories which do not pay attention to case marking in accounting for the CTP predict equal extractability under WH-movement. Our theory, however, predicts that only the genitive NP will be extractable. The facts support our theory:

- (53) a. *Kniga, kotoraja ja xotel, *čtoby* byla na stole ...
 Book, which+NOM I wanted that be+SUBJ on table ...
 b. *Kniga, kotoroj ja xotel, *čtoby* ne bylo na stole ...
 Book, which+GEN I wanted that not be+SUBJ on table

Thus, Russian demonstrates quite conclusively the superiority of a case-related explanation of the CTP over a positional one. This explanation, however, turns out to be something of a two-edged sword. Superficially, at least, Russian is a counterexample to Perlmutter's generalization. We have seen that Russian lacks a dummy element in existential sentences. Similarly in "S-extraposition" sentences:

- (54) Jasno, *čto* vsë v porjadke
 Clear (neuter), that everything (is) in order

Also, definite personal pronouns are not obligatory:

- (55) (Ja) ponimaju, *čto* (ty) imeeš v vidu
 (I) understand what (you) have in mind

These facts should suggest that the NIC is not operative for null NP in Russian, and we should not expect to find the CTP operative. The fact that it is suggests in turn the need for some other explanation of null pronouns in Russian.

This is not unreasonable. Apparent null subjects in Russian differ from their Italian and Spanish counterparts in a number of respects. While the absence of lexical subjects is the norm in Italian and Spanish, where a pronoun is used only for emphasis, the presence of lexical pronouns is the norm in Russian. Pronouns are most commonly dropped only when their referent is extremely close in the discourse, and their presence is in no way marked. Only in the case of existentials and S-extraposition structures are "pronouns" not used; they are, in fact, unavailable, although a variant of sentences like (54) with a pleonastic demonstrative *eto* may be an instance of an actual dummy pronoun (if it is not right-dislocation):

(56) Èto jasno, što vsë v porjadke

Let us suppose that Russian differs from languages like Italian along the following parameter: Italian sentences obligatorily contain all phrase structure positions subcategorized for by the verb; thus, if a sentence lacks a phonetically realized subject, there must be a null subject position (in *Aspects* terms: [NP, S]), which, in turn, must be exempt from the NIC. Russian sentences, however, are not subject to such a requirement; in subjectless sentences, [NP, S] may simply not be present in the phrase structure tree. Thus, no subject node will be present, empty or filled, to be subject to the NIC, and it will be irrelevant to the surface phenomenon of subjectless sentences whether or not null NP falls under that condition. Thus, an Italian sentence like *amo Maria* will differ from a Russian equivalent like *ljublju Mašu* by having a null subject position missing from the Russian:

- (57) a. [_S [e] Tense [_{VP} amo Maria]]
 b. [_S Tense [_{VP} ljublj_u Maš_u]] 'I love Mary'

Thus, in Russian, (57b) is possible even if the NIC does apply to null NP. Now there is one class of cases in which, regardless of whether subjects are required by the base, one must assume the presence of a null subject position in a tensed S. This is, of course, the case of WH-movement and allied phenomena. If we find a nominative WH word in COMP, it must have moved from somewhere, and under trace theory that position must be present in surface structure (S-structure), and will be potentially subject to the NIC. Viewed from the point of view of Logical Form, if a nominative WH-word is interpreted as a quantifier, a nominative trace must be present at its extraction site to serve as a variable for it to bind. Let us suppose that Russian is like English with respect to the NIC, and null NP is subject to the constraint. Thus we see how Russian could show superficially "null" subjects and still exhibit the CTP.

It would be nice if this solution could explain the major difference between Italian and Russian subject pronoun usage: the obligatory emphaticness of the Italian pronoun versus the unmarked status of the Russian pronoun. One possibility might lie in a revision of Chomsky's (forthcoming) "Avoid Pronoun" principle (also exploited by Jaeggli, 1980). The principle, invoked to explain some apparent cases of disjoint reference, requires the avoidance of a lexical pronoun whenever possible. Thus, from this principle it follows that a language like Italian, allowing null pronouns, will assume a marked reading when lexical ones are used. Now assume that the principle is stated almost in reverse: "Use Null Pronoun when Possible". The Italian will follow as before, but the new principle

will say nothing about Russian, where null pronouns are not possible. Russian's truly absent subjects do not fall at all under this new principle, and no marked reading will be predicted for lexical subjects, since they are "unavoidable" under the principle.¹⁷

Another observation which must be made is that the phrase structure absence of a pleonastic NP position in existentials and S-extraposition sentences should not cause any particular problems for interpretation. In the case of extraposition sentences, we do not in any case know that the "extraposed" S may not simply be a postposed subject sentence (*pace* Koster, 1978). In the case of existentials we must merely not assume an "E-rule" of existential interpretation as in Milsark (1974; 1977), triggered by a dummy subject. Some other principle of interpretation must apply, and the problem does not seem insuperable. I leave a solution to further research. (See footnote 13.)

Finally, note that the sort of parametric variation we are proposing to distinguish languages like Italian from Russian makes a certain sense. First of all, not only subjects, but also objects may be omitted in Russian, but not in Italian (or English). Such omission is heavily dependent on context, presumably because no object agreement rule helps clarify a missing object's referent. Thus note the following discourse. Its equivalent is not possible in Italian.¹⁸

(58) – Vy videli novyj fil'm Tarkovskogo?

You saw new film of Tarkovsky

– Da, videl.

Yes saw (singular) 'Yes, I saw it'

Also, one might suggest that these features of Russian are in some way functionally linked to its being an arguably non-configurational (scrambling) language, in opposition to Italian. A child learning a language he discovers to have fairly regular word order might find it reasonable to construct for himself a phrase-structure template, against which utterances may be matched. A missing element in a sentence must then be interpreted as a null element in phrase structure, subject to the conditions of Universal Grammar. For a child learning a non-configurational language, however, such a template would not make sense. Thus, the absence of a subject might not lead to the assumption of any particular null category, under a particular processing strategy. The one exception, again, would be a null category that must be present to serve as a variable bound by an operator, like a WH-word. Certainly, the optionality of certain argument positions in surface structure appears to be a characteristic of many non-configurational languages (e.g. Japanese, as Ken Hale (personal communication) suggests).

In summary, then, we have seen that Russian provides powerful evi-

dence for a non-positional theory of the CTP, and in particular for one that links it to nominative case. What difficulties Russian raises seem to be resolvable with a minimum of additional assumptions, mainly about the obligatoriness of phrase-structure positions. For additional work on this question, see Borer (1980).

3.4. *Conclusions*

In this section I have tried to show that apparent counterexamples to a simple surface statement of Perlmutter's generalization, found in several languages, nonetheless lend themselves to rather simple explanation within the theory I have developed. In each case, I have tried to find some independent motivation for the language-specific assumptions we have needed to make. It is not a fault that I have relied on facts about individual languages independent of the present analysis to get myself out of difficulties. The surface facts of languages do differ, and it is only to be expected that language-particular facts will often get in the way of the expected surface results of universal principles.

4.0. THE CTP AND THE *NP TNS VP FILTER

4.1. *Modern English*

Recalling the discussion of simple relative clauses in section 2.1., I return to a major claim of this analysis: that we can derive not only the CTP, but also the main part of Chomsky & Lasnik's *NP Tns VP filter from the NIC, given our ordering of the components.

I will first suggest, somewhat sketchily, but, I hope, convincingly, that Chomsky & Lasnik's filter, in the form in which they finally state it, collapses several phenomena which should not be collapsed. If their collapsing were to be correct, it would, of course, not contradict our analysis of the CTP, but would make our derivation of a piece of the *NP Tns VP filter otiose, since it does not account for all the phenomena accounted for by the filter.

The piece of the filter which our analysis captures is, of course, the fact that a WH and complementizer cannot both be deleted in COMP when WH has been extracted from the nearest subject position. Thus, the following are ungrammatical in standard English:

- (59) a. *I don't like any man doesn't get up late
b. *The man is a Dohnanyi fan just walked in

As I showed in 2.1., an NP like the object of *like* in (59a) will be ruled out by the NIC in English, since its relative clause complement contains a nominative trace which is left unbound in $\bar{S}$ after deletion:

- (60) [NP any man [$\bar{S}$ [COMP ^e] [S [_{+nom}^{t_i}] doesn't get up late]]]...

Chomsky & Lasnik's filter will, of course, also rule an NP like (60) out. The filter follows deletion and states, in Chomsky & Lasnik's all-encompassing version:

- (61) * [_{α} NP tense VP], unless $\alpha \neq$ NP and is adjacent to and in the domain of [+F], *that*, or NP.

The part of the filter that we have derived is the part that blocks an offending sequence when $\alpha \neq$ NP. There are two other major phenomena which the filter is intended to exclude.

The first is where $\alpha =$ S and the complementizerless $\bar{S}$ is the complement of a [-F] verb or adjective. These include "non-bridge verbs" in the sense of Erteschik (1973)—mostly manner-of-speaking verbs (*mutter*, *quip* etc.)—and a few adjectives like *sad* or *aware*, all of which prohibit *that*-deletion in their complements, which of course are generally of the form *NP tense VP*:

- (62) a. John muttered *(that) he had another appointment
b. I'm sad *(that) Therese is going away

Elsewhere (Pesetsky, 1978), I have suggested a different explanation for this phenomenon, based on the idea, suggested by Bruce Hayes, that these undeletable complementizers are "lexical" in some sense not shared by complementizers with other verbs. Whatever the real explanation may be (also see Appendix 2), it seems rather unlikely that the phenomenon should be handled by a surface filter. Chomsky & Lasnik stipulate, correctly, I believe, that filters should not be allowed to analyze variables. Nonetheless, their filter (61) above will need a variable in a formal version of the *unless*-clause, since *that*-deletion is also blocked when a PP or adverbial material intervenes between an $\bar{S}$ complement and a [-F] verb:

- (63) a. John used to mutter to his friends *(that) he was bored
b. Harry quipped inanely *(that) *dog* is *god* backwards
c. Tim is sad sometimes *(that) Canada is so close to the U.S.

Such a weakening of the theory of filters is presumably not desired.¹⁹ Additional evidence suggesting that these cases should not be collapsed with deletion of subject-bound WH in COMP is the weakness of the effect in (63), in contrast to the clear judgements of native speakers on sentences like (59a–b) (see again Appendix 2).

The second major phenomenon which (61) is intended to exclude, besides examples like (62a–b), is the case of a complementizerless sentence-initial $\bar{S}$, whether a subject or a topicalized object:

- (64) a. *John came today surprised me
 b. *John is a fool he doesn't believe

Here, obviously, α which equals $\bar{S}$, is not adjacent to and in the domain of [+F], *that* (which would make the sentences grammatical), or NP (as in relative clauses with deleted WH). Therefore, the filter assigns star to (64a–b). Chomsky & Lasnik incorporate sentences like (64a–b) into their filter in response to claims by Bever & Langendoen (1971) that this sort of phenomenon is best accounted for functionally, since a sentence-initial $\bar{S}$ without a complementizer will always be mistakenly parsed as a main clause. Bever & Langendoen also try to account for phenomena like those in (59a–b) by a similar principle of avoiding a false parse. In response, and, in fact, following Bever & Langendoen, Chomsky & Lasnik point out that there are dialects of English which do not show the phenomena of (59) (I will return to these shortly), and suggest that the proper explanation for both (59) and (64) lies in the formal mechanism of the filter.

While Chomsky & Lasnik's point about the non-universality of the phenomenon of (59) is well taken, examination of the facts about the phenomenon of (64) suggests that their filter has, in fact, conflated a universal with a non-universal. It appears to be the case that, even in languages or dialects, like Black English, which allow constructions like those of (59), those of (64) are still ruled out. This is, of course, inexplicable if the two phenomena are inextricably linked results of a single surface filter. Clearly the phenomena of (64) are to be accounted for separately. While some sort of universal filter can be devised just to rule out (64), I see no objection to following Bever & Langendoen in this case and blaming false parsing for the badness of sentences like (64). Bever & Langendoen also note that even in dialects, like older stages of English, which allow WH-deletion as in (59), this deletion is much less preferred sentence-initially. This would follow from the false-parsing explanation. Just as an S-initial *that*-clause with omitted *that* will be misparsed as a main clause, so will an S-initial NP followed by a relative clause with empty COMP and extraction from the nearest subject position. Thus both (65a) and (65b) will induce false parses, and are apparently universally disfavored:

- (65) a. *The man came in is my brother
 b. *John is coming today bothers me

This contrasts with a sentence like (66), which, as we shall discuss in the next section, is not universally bad:

- (66) %I knew a man didn't need sleep

This all suggests that Chomsky & Lasnik's filter is an incorrect conflation of disparate phenomena.

Finally, I will cite two arguments suggesting that Chomsky & Lasnik's filter is inadequate even to deal with the phenomenon of (59). The first argument will also apply to their account of other phenomena.

We have seen above that the placement of PPs and adverbials between the sequence *NP Tns VP* and various elements mentioned in the "unless... adjacent to" proviso does not seem to affect the working of the filter, and that these adverbs must be mentioned as a variable in the filter, since they must be placed in their surface position before the filter applies. It was argued that this would entail a much less restrictive theory of filters, and thus make Chomsky & Lasnik's solution less desirable as a whole.

Clearly the same argument applies to the body of the filter. It too must include variables. Thus there is no difference between the grammaticality of (65a) and:

(67) *The man occasionally came in is my brother

This power is necessary for the other phenomena covered by the filter too. There is no grammaticality distinction between (65b) and:

(68) *John is possibly coming today bothers me

Thus we see that Chomsky & Lasnik's theory of filters, to truly explain *NP Tns VP*-phenomena, must be rather more powerful than they suggest. It is clear that the NIC can account for sentences like (67) with no difficulty. An unbound nominative trace is an unbound nominative trace regardless of environment. Similarly, a false-parsing account of (68) (which would presumably label it grammatical but unacceptable) will have no difficulty with the adverbial, since the adverbial does not make its S any less main-clause-like.

The second argument shows an explicitly false prediction of Chomsky & Lasnik's account of sentences like those of (59). Note that deletion of the contents of COMP in sentences like (59) – or, to use traditional terminology, "deletion of a subject relative" – is impossible even when a relative clause is extraposed:

(69) *A man came is my brother

Obviously, this sentence contains no instance of the fatal sequence *NP Tns VP* which could make it subject to the filter. One can, however, account for this in English by claiming that extraposition from NP is a stylistic rule, and thus, in Chomsky & Lasnik's model, follows surface filters. Thus, when filter (61) applies, (69) will still look like (70):

(70) *A man is my brother came in

-- and will be ruled out by the filter.

Standard evidence for the stylisticity of extraposition from NP in English is the fact that extraposed relatives are islands, just like non-extraposed relatives:

(71) *The linguist who a man arrived that knew [e] is Bloomfield

If extraposition is stylistic in English, WH-movement will be forced to apply before it, since WH-movement is syntactic, thus explaining (71) as subadjacency (CNPC) violation.

This solution works for English, but it will not work for Norwegian. Sentences analogous to (71) are fully grammatical in Norwegian, and Taraldsen (1978) has argued convincingly that extraposition from NP must thus be a *syntactic* rule in this language. Only under this assumption will WH-movement be able to apply after extraposition from NP, escaping subadjacency violations in sentences like (71). If extraposition is syntactic, however, it must then precede filters. In this case, we expect that a sentence like (69) will be grammatical in Norwegian. We know that filter (61) is active in Norwegian, if filter it is, since Norwegian sentences like (70) are ungrammatical.²⁰

(72) En man *(som) var fra India kom inn
A man that was from India came in

Som is deletable in other contexts. The extraposed version of (72), with *som* deleted, is just as bad as (70), contrary to the predictions of the filter:

(73) En man kom inn *(som) var fra India

One could, of course, assume another surface filter to rule out (73), but the resulting loss of generalization is obvious. On the other hand, the theory presented in this paper can clearly handle (72) and (73) with no difficulty at all. In each case, the relative clause contains a nominative trace unbound within it:

(74) [$\bar{S}$ [COMP^e] [S [t_i _{+nom}] var fra India]]

Remember that the CTP is found in Norwegian; so trace must be an anaphor for the NIC. And, once more, it is obvious that an unbound nominative trace is unbound regardless of its environment. We correctly predict

that extraposition of an $\bar{S}$, regardless of whether the rule is syntactic or stylistic, will not affect binding relations within it.

4.2. *NP Tns VP in Old English and Black English

The unified account of the CTP and the *NP Tns VP filter presented here makes an interesting prediction not shared by Chomsky & Lasnik's system. If the *NP Tns VP filter results from a nominative trace being left free in $\bar{S}$ and violating the NIC, then the filter should not show up in a language in which null NP is not subject to the NIC. Should this turn out to be the case, it would provide extremely strong support for the present analysis. In the following sections I will briefly sketch some evidence from Old English and Black English suggesting that the generalization may be true, but the data demand further investigation. Note that a large number of languages, like Italian and Dutch, will not furnish the evidence necessary to check this generalization, since they independently disallow empty COMPs. One needs a language which, like English, permits empty COMPs in order to show the generalization. To some extent, Old English appears to have the right properties, and we will consider it first.

4.2.1. Old English²¹

With respect to CTP violations and null subjects, Old English seems to have behaved like Dutch A. A rule like English *there*-insertion, Dutch *er*-insertion or a lexicalist equivalent obtains, as well as an *it*-insertion for extraposed $\bar{S}$'s. Cynthia Allen has informed me that *ther*-insertion seems to have been optional, that is, the pleonastic element may be missing. Thus, sentences equivalent to (75a) occur parallel to sentences like (75b):

- (75) a. There is then on day terror the greatest (= 'There will be on that day the greatest terror')
 b. Was also another miracle (= 'There was ...')

Visser gives a number of examples of $\bar{S}$ -extraposition with a missing expletive (*h/it*).

- (76) a. Wæs geworden þæt Hæland genealæhte Gericho
 Was happened that Savior neared Jericho
 b. Gelamp þæt him ansende sawla Neriend witgan
 Happened that to him/them sent Saul's Savior a prophet+ACC

To explain the occurrence of these null subject positions in Old English, we will assume that, as in Dutch A, null NP is not subject to the NIC. As in Dutch A, we will account for the absence of null *personal* subjects by assuming that Old English simply never developed the required interpretive rule.

If all this is correct, we will predict, of course, that Old English should show violations of the CTP. As Allen (1977, 1980) has noted, it does. I cite two examples from what Allen claims are a large number of cases:

- (77) a. Ac ic wolde witan hu ðe ðuhte be ðaem monnum ðe
 but I would know how thee+DAT seemed the man that
 wit aer cwaedon ðaet unc ðuhte ðaet ___ waeron wildiorum
 we earlier said that us+DAT seemed that ___ were wild-beasts-
 gelicran ðonne monnu (Boeth. 38.5)
 like-er than men
 'But I would like to know how it seemed to you about the men
 that we said earlier that ___ were more like wild beasts than men'
- b. Mine gebroðra, ne lufiga ge ðisne middangeard ðe ge geseo
 My brethren, not love you this world that you see
 ðaet ___ lange wunian ne maeg (Alc. Th. XL)
 that ___ long last not may
 'My brethren, do not love this world, that you see that ___ can
 not last long'

Thus, we find Old English conforming to the general pattern of Dutch A, and allowing the same explanations. We will assume that we have justified the assumption that null NP is not subject to the NIC in Old English.

Unlike Dutch, Old English, at least to some extent, allows empty COMPs in relative clauses. We can thus test our prediction that Old English will lack the constraint covered by the relative clause part of the *NP Tns VP filter. There has been some debate in the philological literature about whether Old English truly allowed "non-introduced" relative clauses at all (see, for example, references on this subject in Mustanoja (1960), pp. 206–8). Visser (1963), however, cites a number of occurrences of this construction, with both missing "object relatives" and missing "subject relatives", though he does remark that missing subject relatives are somewhat less common in Old English. He also criticizes the view that examples of these constructions are to be interpreted as coordinate, rather than subordinate structures (vol. I § 18, 627–8). If missing subject relatives are a real phenomenon in Old English, they provide evidence of the correlation with null subjects and CTP violations that we are looking for. Examples are also provided by Bever & Langendoen (1971), from whom I draw these sentences:

- (78) a. Her on þys geare gefor Ælfred wæs Bapum gerefa
 In this year died Alfred was at Bath reeve
 (Chron. 906)
- b. Sum welig man wæs hæfde summe gerefan
 A rich man was had a steward
 (i.e. There was a rich man who had a steward')

Thus, if the data are correct, Old English shows exactly those effects which the present theory predicts should correlate with the non-applicability of the NIC to null NP, and provides striking evidence for this analysis.²²

4.2.2. *Black English*

I am extremely grateful to Mark Liberman for pointing out the following facts to me.

As has been noted by Labov *et al.* (1968) and Labov (1972), Black English dialects, along with many southern white dialects, typically show omission of subject relatives, violating the *NP Tns VP filter of Chomsky & Lasnik (Chomsky & Lasnik note this):

- (79) a. An tha's the rules mean ²³
 b. I got a man to come out here Saturday is gon' paint
 c. We have very few go to college (Labov, 1972, p. 189)

Let us suppose that these sentences show that null NP in Black English is not an anaphor for the NIC, as we assumed in Dutch A and Old English. If this is so, then Black English should show violations of the CTP. Examples that would show this are not to be found in any corpuses which I have been able to examine. In fact, these corpuses show hardly any extractions from embedded clauses, and I have found no examples of extraction from subject position. An example is heard, however, in the following, presumably impromptu, sentence from the introduction to a blues song on a recording by Junior Wells (*Chicago, the Blues*, Vol. 1). He is about to sing a song a friend wrote:

- (80) It's a number that he made that I think that ___ will linger in your hearts forever

In the absence of further transcribed examples, one might consider this to be a slip. However, Wilson Gray (personal communication) suggests that sentences like (80) are generally grammatical in Black English. Interestingly, his own speech, though otherwise heavily influenced by Standard English, lacks the CTP altogether. In the absence of other data, we cannot be sure that Black English does violate the CTP; still, if further investigation confirms our hypothesis that it does, then the correlation with violations of the *NP Tns VP filter will provide interesting evidence for the present analysis.

I have not, of course, spoken about the third phenomenon our theory predicts should be found in Black English, namely, null subjects. To be sure, these do not occur freely, as in Italian or even as in Dutch A. Still, one may recall the discussions in Labov (1972, pp. 188–9) and Labov *et al.* (1968, pp. 283–7) of sentences like:

- (81) a. Ain't nothin' you can do about it
 b. Ain't nothin' went down

Labov and his colleagues note that (81a–b) do not admit of an analysis in terms of subject-AUX inversion (which is regular in Black English negative sentences) and that they suggest an analysis with a missing dummy subject (Black English *it*). If this is right, we have further support for our analysis in the correlation between CTP violations, *NP *Tns* VP violations and null subjects. The problem is to explain the limitations on the distribution of these null subjects. In particular, they seem possible in negative sentences, but not in positive sentences. We must therefore leave our analysis of Black English somewhat open.

4.3. *Conclusions*

While I have noted the uncertainty of the data, this section has attempted to show that our theory can account for an apparently valid correlation between phenomena that on the surface appear to be unrelated. If correct, this provides the strongest support for our analysis, particularly since no such correlation is predicted by other reasonable theories of the individual phenomena, such as the system of filters proposed by Chomsky & Lasnik, or Bresnan's Complementizer Constraint.²⁴ In general, I take it as an advantage of the analysis I am proposing that it makes a large range of predictions and attempts to account for more than a single surface phenomenon. If we are to construct a maximally explanatory theory of syntactic structure, we will hope that apparent complexities of individual languages will turn out to be accountable for as predictable results of interactions between simple, independently motivated mechanisms. Such a theory will be inherently more falsifiable than one which accounts for individual phenomena by individual mechanisms, but will be the more interesting in its falsifiability, since its potential for making testable predictions about unexamined data will be greater.

5.0 PREPOSITIONAL PHRASES AND THE *FOR* COMPLEMENTIZER

In this final section I will briefly consider two points raised by Bresnan (1977) which create some problems for this analysis. The first involves extraction of PPs (as well as some adverbial WH-words) from embedded sentences, which can produce effects suggesting the CTP; the second point involves the status of extractions from subject position in *for*-complements, which also seem to produce effects suggesting the CTP, at least in Standard English. In the first case we will suggest a way the phenomenon in question can be integrated into our theory; in the second case we will suggest that the phenomenon might not be so integrable.

5.1. *PPs and Adverbial WH*

Bresnan notes that some types of fronting motivate an inversion of subject and VP:

- (82) In these villages can be found the best examples of this cuisine
(= Bresnan (39))

Bresnan shows that extraction of the PP from this sort of construction can produce a violation of the CTP in an embedded sentence. Extraction of an adverbial WH-word, we note, can have the same effect. Her examples are from cleft sentences; we shall use ordinary relative clauses:

- (83) a. These villages, in which we all believe (*that) _____ can be found
the best examples of this cuisine, are in the South
b. These villages, where we all believe (*that) _____ can be found
the best examples of this cuisine, are in the South

To explain these examples, our theory must assume that the fronted PP or adverbial has moved into subject position ([NP^{PP}]) before WH-movement. If nominative case assignment applies to the whole subject NP position, subsequent WH-movement from that position will leave a nominative trace ([NP +_{nom}^e]) subject to the NIC as discussed in this paper. The CTP effects will follow in the normal fashion.

As Joan Bresnan and Donca Steriade have pointed out to me, the idea that these PPs and adverbials move to subject position is supported by the fact, noted by Postal (1977) that these elements can undergo Subject-to-Subject raising:

- (84) In these villages seem to be found the best examples of this cuisine.

Thus, this construction, which Bresnan took to be an argument for her Complementizer Condition (preventing leftward movement over a variable ending in COMP), poses few problems for our analysis of the CTP, since any theory will need to consider some fronted PPs and adverbials as subjects, to explain the Raising facts.

5.2. *For-Complements*

Chomsky (1980) discusses *for*-complements in English, and motivates an explanation, in terms of case-marking conventions, for the obligatory presence of *for* in sentences like:

- (85) a. It is preferred for John not to see Mr. Murdoch
 b. I want very much for John to like his roommate

Extraction of the subject of a *for*-complement appears to be blocked in Standard English, an effect reminiscent of the CTP:

- (86) a. *John's the one guy it is preferred for _____ not to see Mr. Murdoch
 b. *John's the person who I want very much for _____ to like his roommate

For unknown reasons, *for* can be deleted in (86b), yielding a somewhat better sentence, while it cannot be deleted in (86a). The improvement in (86b), occasioned by the deletion of the complementizer, all the more suggests the CTP. Nonetheless, our theory will clearly not account for (86a–b), since the extracted subjects are not nominative, and will not be subject to the NIC. According to Chomsky & Lasnik (who had no stake in this point), this is a welcome result, although a few of their arguments are not applicable here. Their main point is that, while the CTP, for them, is a universal, with some explainable exceptions, the phenomenon of (86) is dialect-specific. They cite the possibility of structures like (87) in Ozark English:

- (87) who are you going to try for [e] to VP (= C&L (213))

Let us suppose that Ozark English does not pattern with Black English in generally allowing violations of the CTP. Chomsky & Lasnik do not discuss this point, and I have not been able to obtain relevant data. (They also cite some examples like (87) from Dutch, which Maling & Zaenen (1978) reanalyze.)

Clearly our analysis suggests that Chomsky & Lasnik's conclusions are correct. There remain, however, some problems. They propose to rule out sentences like (86a–b) by a special, dialect-specific filter:

- (88) *[for - to] (= C&L (61))

Unfortunately, this filter, like their **NP Tns VP* filter, which we rejected above, must be reformulated to allow for various types of intervening material, like *not* in (86a), or an adverbial:

- (89) *You're someone who I want very much for really to have fun

A number of solutions are available, but the problem must be noted. (Also see Bresnan (1977), p. 195, note 12 for further problematic examples.)

6.0 THE CTP AND LINGUISTIC THEORY

Much of this paper has been an implicit and explicit comparison of our theory of the CTP with other proposals. In a sense, NIC-centered theories of the phenomenon represent a return to Perlmutter's original ideas about the constraint -- that it is crucially dependent on the notion "subject", or in the given case, the notion "subject of a tensed sentence". In the system of Chomsky (1980), it is the subject, [NP, S] which receives nominative case from tensed Tns-node.

The subject-oriented approach to the CTP stands in contrast to the approach of Chomsky & Lasnik and of Bresnan (1977), where adjacency to COMP of the site of extraction is taken to be the determining factor for the CTP. Their accounts differ in that Bresnan attempts to constrain the actual application of a transformation to a position adjacent to COMP, while Chomsky & Lasnik rule out the presence of a trace resulting from such an application in this position. The theory of Talralsen (1980) is a mixture of an NIC approach and an adjacency approach, combining the NIC with a filter on string-vacuous movement, in the spirit of Chomsky (1973), along with a [complementizer WH] order in COMP to derive the appropriate effects. Evidence from Russian, as discussed in section 3.3 above, seems quite deadly to the various adjacency theories, since there it can be shown that scrambling, a syntactic process preceding semantic interpretation, whether basal or transformational, does not affect the appearance of the CTP, while it clearly changes string adjacency relationships. This difficulty can, of course, be avoided in various ways, by complicated reorderings of the components of the grammar, but these all seem to involve unnecessary complications. One can similarly note that, even in English, the CTP is not affected by the presence of other material between subject position and COMP. Extraction from embedded clauses in which material has been fronted is generally somewhat odd in English, but I believe there is a contrast between (90a) and (90b);

- (90) a. *John, who I'm sure that in July said something different from what he's saying now, is a big liar
 b. ?John, who I'm sure in July said something different from what he's saying now, is a big liar

It seems reasonably clear that adjacency to COMP is irrelevant to the CTP.²⁵

In general, while problems remain, it seems that the present theory also accounts more naturally for cross-language variation with respect to the CTP and attendant phenomena. This is not because the theory appears to be any less restrictive, but because it pinpoints in a reasonably economical manner the areas in which languages may be expected to differ. Further-

more, the present theory carries with it the bonus of an explanation for the non-deletability of elements in COMP with a nominative extraction site in the nearest S, the phenomenon accounted for by Chomsky & Lasnik's **NP Tns VP* filter, and predicts a correlation between the non-occurrence of this effect and the absence of the CTP, which seems to be correct. Our account of the **NP Tns VP* filter also eliminates certain serious empirical difficulties with the filter account, involving extraposition and adverb placement.

This subsuming of several surface phenomena under a single deeper explanation I take to be the major obvious advantage of the present proposal. Every theory is built on some core of arbitrary stipulation, but it is always desirable to minimize the number of stipulations. If such a reduction is correct in a given case, it is also likely that it will have desirable empirical consequences, and will do better in explaining hitherto unexplained facts.

In particular, this theory, combined with Chomsky's (1980) elimination of the **NP to VP* filter, seems to cast serious doubt on the entire concept of universal syntactic filters as proposed by Chomsky & Lasnik. First of all, although a number of phenomena are collapsed in several of their filters (at the cost of severe complications in their statement), their filters remain pretty much an ad hoc series of descriptions of various undesirable surface configurations, on a one phenomenon/one account basis. In the absence of a better explanation of these phenomena, such an account cannot, of course, be faulted. On the other hand, if the present theory is correct, it provides just such a better explanation, relying almost exclusively on characteristics of the theory of Universal Grammar independently needed. Second of all, it is implausible, as the Russian examples show, to think that Universal Grammar will contain stipulations regarding the surface order of constituents, i.e. syntactic filters, when languages differ to such a great degree in characteristic word orders.

On the other hand, it is striking that one of Chomsky & Lasnik's main contentions turns out to be true, perhaps in a deeper way than they realized. A basic point of their paper was that phenomena which had been taken to motivate such artifices as extrinsic ordering, obligatory applications of rules and arbitrary conditions on their application could be accounted for, especially within trace theory, by conditions on surface structure. The NIC, like their filters, is clearly such a condition, but represents a surface constraint on *binding relations* (a Logical Form constraint, perhaps) rather than a condition, or set of conditions, on syntactic structure.

Surface structure constraints on syntactic strings may exist, like the clitic-ordering constraint motivated for Spanish by Perlmutter, or, perhaps, the **[for-to]* filter discussed above. We have obviously leaned heavily on a filter against doubly filled COMPs in the body of this paper,

although other possibilities are available (see Appendix 1). There is little evidence, however, that such constraints should be part of "core grammar", and seem to be language specific. Given their peripheral character and their high tendency to inter-dialect variability, one may suppose that they are not universal and may be learned by explicit teaching, like irregular verbs or the preferability of *John and me* to *me and John*.

"Real" surface filters thus may be arbitrary, although other conditions may govern them. The explanation for constraints like the CTP, on the other hand, arises from deeper principles, if this paper, and others like it, are correct. Note that if Chomsky & Lasnik's filter were correct, there is no reason why it should specify $\pm WH$ as a conditioning environment. It could as well be $+WH$, $-WH$ or any other element. Similarly, to explain the problem of the *que/qui* alternation in French, Chomsky & Lasnik invoke some mysterious feature which makes *qui* differ from $\pm WH$ and renders it immune to the filter. Manifestly, a similar feature could be invoked to make any lexical item immune to any filter, rendering a filter's claim to explanatory value very weak. Such excessive power is not available in the system I have proposed here, which offers an explanation of the very nature of *qui* which, while at present not overelegant, accounts automatically for its exceptionality with regard to the CTP, without any mysterious invocation of unspecified features.

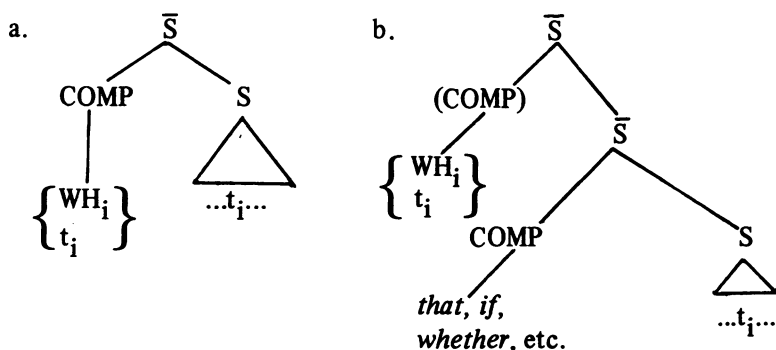
While many features of the system are still unclear, and many phenomena are underexplained or unexplained, I believe the evidence of this paper, along with Chomsky (1980), strongly suggests the important role played by the subject of a tensed sentence, a nominative NP, in universal grammar, and by the binding relations between this subject position and other elements in the sentence. What this unique role may result from is a matter for further work.

APPENDIX I: DOUBLY-FILLED COMPS OR ADJUNCTION TO $\bar{S}$?

The device of doubly filled COMPs, so important in the above analysis, has been argued against by Bresnan (1976) and others, on the basis of evidence from Swedish and other languages, which may suggest that a WH and complementizer, when allowed to surface together, do not form a constituent. I will not review the arguments here. Let us imagine that they are correct, and that COMP does not branch, nor are moved elements adjoined to it. We can still construct a proposal with a number of the advantages of the one presented above.

Suppose that there is no rule deleting complementizers like *that*, but that the optionality of lexical insertion accounts for all alternations of complementizer with zero. Let us imagine that WH-movement can still land in COMP. It will follow that such a landing is only possible if no complementizer has been inserted in that position. If COMP is filled, how-

ever, let us assume that the moved element can Chomsky-adjoin to $\bar{S}$. This makes some sense as movement to COMP if COMP is defined simply as a left-most non-argument position in $\bar{S}$. This last operation, however, will leave its trace unbound in the lower $\bar{S}$. The two possible outcomes are shown below:



It is obvious that if the trace in (b) is nominative, it will be in violation of the NIC. Thus, this theory yields an explanation much like the one presented in the text for the CTP.

One crucial difference between the two theories lies in the parameter of doubly-filled COMPs. Clearly the theory just sketched will need to rule out (b) as a possible outcome of WH-movement in languages like English, while perhaps allowing it in languages like Swedish or Middle English, that allow constructions like this to surface. Let us assume some sort of filter accomplishes this. The point is, this new theory predicts no difference in the presence of CTP depending on this parameter. Whether or not a language allows constructions like (b) to surface, any application of WH-movement out of a clause headed by a complementizer will leave the trace of WH in S unbound within $\bar{S}$. The theory presented in the text, however, predicts that, if languages like Swedish actually do not have the filter against doubly-filled COMPs, they will also lack the CTP. This prediction appears to be false (cf. Taraldsen (1980) for some discussion of Norwegian data, and particularly Bennis (1980) for some discussion of Dutch). Thus there is some evidence supporting a theory like the one just presented, which does not rely on the filter (14), over the theory sketched in the body of the paper. This argument, however, rests entirely on the assumption that languages permitting "doubly-filled COMP" phenomena actually show structures like the one blocked by (14). The arguments in Bresnan (1976) and elsewhere might be used to show that, in those cases only, such a structure does not obtain, and filter (14) holds. The problem awaits further investigation.

APPENDIX II: EXTRACTION FROM ADJECTIVE COMPLEMENTS

Kayne (1980) presents an interesting alternative explanation of the CTP, also in terms of case-marking and the NIC. I will discuss his proposal briefly. While it is extremely interesting and potentially not incompatible with certain other proposals suggested in this paper, it hinges on some data involving extraction from the complements of adjectives which I believe to be suspect.

He assumes WH-movement into COMP, and the mechanism of doubly filled COMPs assumed in the body of the present paper. His essential proposal is to modify the notion "free" in the NIC to "case-free". An NP is "case-free" if it is not bound by a *case-marked* antecedent. Additionally, he suggests that a trace in COMP can be marked for case (and thus bind a nominative anaphor in $\bar{S}$ under the NIC) by a matrix verb. This case marking is prevented, however, by a branching category boundary intervening between verb and trace, an assumption Kayne claims is independently desired. In the cases of a CTP violation, where a complementizer *that* has been base-generated, COMP will branch, and a trace in COMP binding a nominative trace in S will not be case-marked by the matrix verb. The resulting violation of the NIC will rule such a construction out. If no complementizer is generated in COMP, however, then a trace will be able to be case-marked and bind a nominative anaphor under the NIC, preventing an NIC violation.

Apart from the theoretical difficulties in such a proposal—why case-marking should be allowed to apply to elements in COMP in the case of all verbs, while case-marking into subject position of an embedded S in Chomsky (1980) (which Kayne cites as a precedent) should be so restricted; and why branching should rule out case marking in COMP and not, say, in conjoined NPs—there are empirical problems with Kayne's analysis.

The most interesting evidence Kayne gives in support of his view comes from adjectives. He notes that adjectives in English never take NP objects (*worth* and *near* perhaps being exceptions). He suggests that this is due to the fact that adjectives do not assign objective case, combined with a filter on caseless lexical NPs. If adjectives do not assign case, we will expect them not to assign case to traces in COMP. Thus, if a trace in COMP must be case-marked in order to bind a nominative trace in S under the NIC, no nominative traces should be possible in the complements of adjectives, whether a complementizer is present or not. Extraction of the subject of a tensed complement to an adjective should always violate the NIC. Kayne, in fact, suggests that this is true, citing examples like the following:

- (91) a. John, who it's essential she talk to...
 b. *John, who it's essential talk to her...
 c. John, who it's not obvious anybody can stand...
 d. *John, who it's not obvious gets along with anybody...

(Examples with topicalization show a sharper contrast.)

He also suggests that verbs which do not assign case, like *seem* show the same contrast:

- (92) a. French, which it doesn't seem Mary knows at all...
 b. *Mary, who it doesn't seem knows French at all...

Kayne admits that the judgments "*" vs. "OK" "may perhaps be overstated", but maintains that the contrasts are real. In fact, it is the very weakness of the effect that suggests that it is not to be grouped with the CTP. Many informants do not get Kayne's contrasts at all, but many who do (like myself) find these contrasts missing with other adjectives:

- (93) a. Harry, who I'm not sure anybody likes...
 b. Harry, who I'm not sure likes anyone...
 c. One man who I'm not happy you invited...
 d. One man who I'm not happy showed up...

I do not have an explanation for this effect which is very satisfactory. However, note what examples like (93 b,d) do to Kayne's theory. The only way he could account for these examples would be to assume that *sure* and *happy* could assign case. But this is not true under his other assumptions, since these adjectives still cannot take direct objects. Clearly, case-marking is not at issue.

What is at issue? Noam Chomsky has pointed out to me that the relative difficulty of extracting the subject of adjective complements seems to be related to the deletability of the complementizer. There seems to be a slight but perceptible difference in acceptability between *that*-deletion with the matrix adjectives in (91) and those in (93):

- (94) a. ?It's essential John talk to her
 b. ?It's not obvious John can get along with anybody
 c. I'm not sure Harry likes anyone
 d. I'm not happy you invited him

In fact, the very negation of the matrix verb, which Kayne uses (along with a negative polarity item in the embedded clause) to ensure a non-parenthetical reading, makes *that*-deletion more difficult and extraction of the embedded subject worse. Running the risk of a parenthetical interpretation in the WH-movement case, compare (95) with (91b) and (94b):

- (95) a. It's obvious John can get along with everybody
 b. John, who it's obvious gets along with everybody...

These observations recall Erteschik (1973), who noted that matrix verbs with "asserted" content tend to prevent *that* deletion in their complements and to make extraction difficult from those complements. It may be that when *that* is deleted in the complements of adjectives with "asserted" content, it is not by the usual process of deletion in COMP (which we have argued must precede semantic interpretation and the NIC), but by a late rule of ellipsis. This would explain the CTP violations, if CTP violations they are, in (91b,d), since a complementizer would be present in the embedded clause at the time the NIC applied, and a trace binding a nominative extraction side would, of necessity, have been deleted.

NOTES

1. Throughout the paper, I will be ignoring the CTP as it shows up with topicalization. If topicalization is WH-movement, as suggested in Chomsky (1977), then our analysis of the CTP will extend naturally to this case.

2. Similar observations are made by Comrie (1973) and Chvany (1975). Thanks to my many Russian informants for these data.

3. Dominique Sportiche has pointed out that, for many speakers, extraction of direct objects from WH-islands is somewhat worse than extraction of PPs, and that such extractions in general are often better with relative clauses than with questions:

- a. Voilà l'homme à qui je savais quelle lettre Marie avait envoyée
Here's the man to whom I knew which letter Mary had sent
- b. *Voilà l'homme que je savais quelle lettre avait écrit
Here's the man who I knew which letter had written

4. See Chomsky (1981) for some solutions to this problem, based in part on Friedin & Lasnik (1979).

5. Note that (25) must follow WH-movement, which is why I have considered it a special case of deletion in COMP, but see below. Like that rule, it must be prevented from applying in indirect and direct questions.

6. Of course, several complementizers work like *that*, for example, *as* (Such men as know John are probably impressed.), which does not differ from *that* with respect to the familiar constraints.

7. I assume that the complementizer *qui* discussed below is distinct from the interrogative pronoun *qui* found in questions.

8. Kayne (1979) cites the subject/object asymmetry in the following examples as evidence that the *que/qui* alternation can occur in structures of control, where no WH-movement has taken place. The Specified Subject Condition would rule out a non-trace null NP, i.e. PRO, in an object position:

- a. Je l'ai rencontrée *que / qui sortait du cinéma
I met her coming out of the movie theater. (lit. '...that came...')
- b. Je l'ai vue *que / qui Jean embrassait
I saw her that John was kissing

Sportiche (personal communication) has suggested that these may be cases of PRO-movement into COMP, based on independent considerations. Trace of PRO in object position will be ruled out by the SSC (Opacity), while the *que/qui* rule might be restated as:

[WH, PRO, t_i que] $\rightarrow$ [qui_i] / etc.

Note that the collapsing of COMP coindexing with deletion, in the form of "COMP Contraction", may be incorrect, as Hans Obenauer has pointed out to me. Certain French dialects fail to delete the complementizer in embedded questions:

- c. Je ne sais pas quand que Jean est arrivé
I don't know when that John arrived

Extraction from subject position can still yield a *qui* complementizer, without deletion:

- d. Je ne sais pas qui qui est arrivé

(Thanks to Anne Rochette for the data.)

9. Bennis (1980) claims that this prediction is not borne out, at least as far as the distribution of Dutch dialects with doubly filled COMPs is concerned. If this is true, it provides evidence for the alternative theory of Appendix I.

10. I follow Kayne (1980) here. See also Taraldsen (1980) for a suggestion involving binding to verbal inflection.

11. If we believe, with Chomsky, that a constraint cannot be learned from negative evidence, we are left with a small puzzle. Since a child will not learn anything from *not* hearing violations of the CTP, we must assume that the unmarked case is for null NP to be an anaphor under the NIC. In other words, a child will assume that the CTP holds, and that null subjects are impossible, unless shown otherwise by positive evidence. Nonetheless, Haiman (1974) notes that languages following the pattern of Italian and Spanish are much more common than languages following the pattern of English and French. If this observation is correct, and it is unclear how it could be meaningfully confirmed or disproved, then we are forced to say that the majority of languages display a marked characteristic. In a sophisticated markedness theory, statistics of this sort may be rather irrelevant in determining marked members of oppositions, since the theory will base itself on a theory of acquisition, but observations like Haiman's are still worth keeping in mind.

12. Thanks to Annie Zaenen and to Joan Maling for discussion of this section and for some data not in their paper.

13. The existential *construction*, of course, requires some special interpretation; the point is simply that no interpretation is necessary of subject position *per se*. See Maling & Zaenen (1978) for a similar conclusion, and Milsark (1974; 1977), for an opposing one.

14. Maling & Zaenen propose that examples like (40)b do not show a null subject position, but rather fronting of the adjective, *duidelijk*, into that position. This analysis is, whatever its merits, impossible for sentence like (40)a, since bare verbs can never be so fronted in subordinate clauses, cf:

- a. *toen bleek het, dat ...
 when appeared it that ...
b. *toen vertelde Jan ...
 when said Jan ...

15. A way out for Chomsky & Lasnik would be to assume the following ordering:

1. Syntax (WH-movement)
2. Filters
3. Scrambling
- 4b. Semantic interpretation:
 - Q-scope
 - etc.

Why scrambling should occupy a special place in universal grammar is, however, unclear. The fact that stylistic rules do not generally seem to affect logical properties of sentences led Chomsky & Lasnik to assume that semantic interpretation could precede stylistic rules. If this position is maintained, we could not claim that scrambling is a stylistic rule.

16. Babby (1980), following some earlier researchers, claims that positive existentials have fixed verb-subject order, while negative existentials have free order. My work with informants has not yielded evidence of such an asymmetry; my informants seem to find any order acceptable, with verb-subject somewhat preferred in both cases. The point is, however, immaterial to our argument. Were Babby correct, we would, if anything, have stronger evidence against COMP-adjacency theories of the CTP.

17. Under this reinterpretation of the principle, we may wish to restate it in more general terms as "Say as little as possible"!

18. Cf. Italian:

–Si, *(l')ho veduto

(Thanks to Giuseppe Longobardi for the datum.)

19. Note that, at least within the framework of Chomsky (1980), it is not possible that the PPs and adverbs in (74) are placed in their positions after the application of surface filters, by a late stylistic rule. Chomsky accounts for the undeletability of *for* in a sentence like:

I want sometimes *(for) John not to talk shop at home

by claiming that the presence of the adverb *sometimes* between the verb *want* and the lower subject prevents objective case marking (defined on adjacency) from being given to the lower subject by the higher verb. The presence of *for* gives the lower subject oblique case. A filter against caseless NPs rules out sentences like this with *for* deleted. The clear presupposition of this analysis is that adverb placement as in (63) is done before case marking, presumably by base generation. Thus we cannot argue that the PPs and adverbs are not in their surface positions in (63) when Chomsky & Lasnik's filter (61) would apply.

Of course, one could alter Chomsky & Lasnik's grammar so that adverb placement follows filters, and case marking follows adverb placement, but such an assumption of late case marking would require some magical mechanism to assign to moved constituents, like WH, the case of their original trace. This is not at present necessary.

20. I am grateful to Tarald Taraldsen for the Norwegian data.

21. Thanks to Wayne O'Neil, Cynthia Allen and Joan Maling for discussion of this section. I have used only secondary sources here; hence my choice of orthography or use of modern "translation" reflects them. The material obviously deserves a closer study, which is beyond the scope of this paper.

22. It is an extremely interesting question how these constructions developed in later stages of English. Middle English apparently shows fewer examples of null subjects, and many more examples of deleted subject relatives, as in (78). I do not know to what extent CTP violations can be found in Middle English. An intriguing example from Chaucer is cited by Jespersen (1961, IV.10.7.3 – he is concerned with the case of *whom*):

... yet wol we us avyse

Whom that we wol that shal ben our justice (CT B665)

Jespersen notes that some manuscripts omit *that*. It is striking that this omission disrupts the meter. Can this date the disappearance of CTP violations from standard English? Of course, Middle English data predict that the CTP will be violated, since apparent cases of doubly filled COMPs are so common (cf. Keyser, 1975). Deletion of subject relatives (or apparent cases of this) survive to the present day in some dialects, but seem limited in occurrence to cleft and existential sentences, where they may represent a different structure (cf. Jespersen III.7.5.3).

23. Note the WH-deletion in an embedded question. Also examples like (79b) show that the phenomenon is not limited to cleft and existential sentences (see preceding footnote).

24. Taraldsen (1980) offers an account of the **NP Ths VP* filter that might make some of these predictions.

25. Chomsky & Lasnik's filter, though not the other theories, is also empirically inadequate in the case of embedded Stylistic Inversion in French, as Taraldsen (1980) has noted. Assuming with Kayne & Pollock (1978), that Stylistic Inversion is Rightward NP-movement, the following sentence, with trace indicated, should violate Chomsky & Lasnik's filter:

Où crois-tu que/*qui t_i va habiter [ton père]_i ?

Under the present analysis, if the postposed NP c-commands its trace, then the nominative trace will not be free in $\bar{S}$, and the NIC will not be violated. The non-application of the *que/qui* rule follows from the analysis in section 2.3, since there has been no movement into COMP.

REFERENCES

- Allen, C.L. (1977) *Topics in Diachronic English Syntax*, unpublished Doctoral dissertation, U. Mass., Amherst, MA.
- Allen, C.L. (1980) "Movement and Deletion in Old English," *Linguistic Inquiry* 11, 261–324.
- Babby, L.H. (1978) "Negation and Subject Case Selection in Existential Sentences: Evidence from Russian," distributed by Indiana University Linguistics Club, Bloomington, IN.
- Babby, L.H. (1980) *Existential Sentences and Negation in Russian*, Karoma Publishers, Ann Arbor, Michigan.
- Bennis, H. (1980) "Coindexing and Complementizer-Trace Phenomena," paper presented at 5th GLOW Colloquium, Nijmegen, Netherlands.
- Bever, T.G. and D.T. Langendoen (1971) "A Dynamic Model of the Evolution of Language," *Linguistic Inquiry* 2, 433–463.

- Borer, H. (1980) "Empty Subjects and Constraints on Thematic Relations," in *Proceedings of the X NELS Conference*, Ottawa, Canada.
- Bresnan, J.W. (1976) "Evidence for a Theory of Unbounded Transformations," *Linguistic Analysis*, 2, 353-393.
- Bresnan, J.W. (1977) "Variables in the Theory of Transformations," in Culicover *et al.* (1977).
- Chomsky, N. (1965) *Aspects of the Theory of Syntax*, MIT Press, Cambridge, MA.
- Chomsky, N. (1973) "Conditions on Transformations," in S.R. Anderson and P. Kiparsky (eds.), *A Festschrift for Morris Halle*, Holt, Rinehart and Winston, New York.
- Chomsky, N. (1977) "On WH-Movement," in Culicover *et al.* (1977).
- Chomsky, N. (1980) "On Binding," *Linguistic Inquiry*, 11, 1-46.
- Chomsky, N. (1981) *Lectures on Government and Binding*, Foris, Dordrecht.
- Chomsky, N. and H. Lasnik (1977) "Filters and Control," *Linguistic Inquiry*, 8, 425-504.
- Chvany, C.V. (1975) *On the Syntax of BE-Sentences in Russian*, Slavica Publishers, Cambridge, MA.
- Comrie, B. (1973) "Clause Structure and Movement Constraints in Russian," in C. Corum, T.C. Smith-Stark and A. Weiser (eds.), *You Take the High Node and I'll Take the Low Node: Papers from the Comparative Syntax Festival*, Chicago Linguistic Society, Chicago, IL.
- Culicover, P.W., T. Wasow and A. Akmajian (eds.), *Formal Syntax*, Academic Press, New York.
- Erteschik, N. (Shir) (1973) *On the Nature of Island Constraints*, unpublished Doctoral dissertation, MIT, Cambridge, MA.
- Freidin, R. and H. Lasnik (1981) "Disjoint Reference and Wh-Trace," *Linguistic Inquiry* 12, 39-53.
- Haiman, J. (1974) *Targets and Syntactic Change*, Mouton, the Hague.
- Jaeggli, O. (1980) *On Some Phonologically Null Elements in Syntax*, unpublished Doctoral dissertation, MIT, Cambridge, MA (to appear, Foris Publications, Netherlands).
- Jespersen, O. (1961) *A Modern English Grammar on Historical Principles*, Allen and Unwin, London.
- Kayne, R. (1979) "Binding, Clitic Placement and Leftward Quantifier Movement," *Rivista di Grammatica Generativa*.
- Kayne, R. (1980) "Extensions of Binding and Case-Marking," *Linguistic Inquiry*, 11, 75-96.
- Kayne, R. and J.-Y. Pollock (1978) "Stylistic Inversion, Successive Cyclicity and Move NP in French," *Linguistic Inquiry*, 9, 595-621.
- Keyser, S.J. (1975) "A Partial History of the Relative Clause in English," in J. Grimshaw (ed.), *Papers in the History and Structure of English*, U Mass., Amherst, MA.
- Koster, J. (1978) "Why Subject Sentences Don't Exist," in S.J. Keyser (ed.) *Recent Transformational Studies in European Languages*, *Linguistic Inquiry* Monograph 2, MIT Press, Cambridge, MA.
- Labov, W. (1972) *Language in the Inner City*, U. of Pennsylvania Press, Philadelphia, PA.
- Labov, W., P. Cohen, C. Robins, and J. Lewis (1968) *A Study of the Non-Standard English of Negro and Puerto-Rican Speakers in New York City*, unpublished ms., Columbia University, New York.
- Maling, J. and A. Zaenen (1978) "On the Non-Universality of a Surface Filter" *Linguistic Inquiry*, 9, 475-497.

- Milsark, G. (1974) *Existential Sentences in English*, unpublished Doctoral dissertation, MIT, Cambridge, MA.
- Milsark, G. (1977) "Towards an Explanation of Certain Peculiarities of the Existential Construction in English," *Linguistic Analysis*, 3, 1-29.
- Mustanoja, T.F. (1960) *A middle English Syntax*, Societe Neophilologique, Helsinki.
- Perlmutter, D.M. (1971) *Deep and Surface Constraints in Syntax*, Holt Rinehart and Winston, New York.
- Pesetsky, D. (1978) "Some Conditions on Bounding Nodes," unpublished ms., MIT, Cambridge, MA.
- Pesetsky, D. (1979) "Complementizer-Trace Phenomena and the Nominative Island Condition," in *Proceedings of the IX NELS Conference: CUNYForum 6*, Queens College Press, Flushing, NY.
- Pesetsky, D. and M. Sinicyn (forthcoming) "AUX and the Genitive of Negation in Russian".
- Phinney, M. (1980) "The Acquisition of Embedded Sentences and the Nominative Island Constraint," paper presented at the XI NELS Conference, to appear in proceedings.
- Postal, P. (1977) "About a 'Nonargument' for Raising," *Linguistic Inquiry*, 8, 141-154.
- Rizzi, L. (1978) "Violations of the Wh-Island Constraint in Italian and the Subjacency Condition," *Montreal Working Papers in Linguistics*, Montreal.
- Rozental', D.E. (1970) *Russkij Jazyk: Posobie dlja Postupajuščix v Vuzy*, Vyšaja Škola, Moscow, USSR.
- Sportiche, D. (1981) "Bounding Nodes in French," *The Linguistic Review* 1, 219-246.
- Taraldsen, K.T. (1978) "The Theoretical Interpretation of a Class of 'Marked' Extractions," [in A. Belletti, L. Brandi, and L. Rizzi (1981) *Theory of Markedness in Generative Grammar*. Scuola Normale Superiore, Pisa].
- Taraldsen, K.T. (1980) "On the Nominative Island Condition, Vacuous Application and the *That*-Trace Filter," Indiana University Linguistics Club, Bloomington, IA.
- Timberlake, A. (1975) "Hierarchies in the Genitive of Negation," *Slavic and East European Journal*, 19, 123-38.
- Visser, F.T. (1963) *An historical Syntax of the English Language*, E. J. Brill, Leiden, Netherlands.
- Woolford, E. (1978) "The Recoverability Principle for Deletion: A Case of Unbound Traces," paper presented at LSA Winter Meeting.
- Zaliznjak, A.A. and E.V. Padučeva (1979) "Sintaksičeskie Svojstva Mestoimenija kotoryj," in T.M. Nikolaeva (ed.), *Kategorija Opredelënnosti-Neopredelënnosti v Slavjanskix i Balkanskix Jazykax*, Nauka, Moscow, USSR.

M.I.T.

Dept. of Linguistics

20C - 128 Cambridge

MA 02139

U.S.A.

